Software Requirements Specification

for

Virtual Whiteboard

A Real-Time Collaborative Drawing and Brainstorming Platform

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the **Virtual Whiteboard**, a web-based application that lets multiple users draw, write, and share ideas on a shared digital canvas in real time. It is intended for developers, testers, project managers, UI/UX designers, and stakeholders, and serves as the baseline agreement between the customer and the development team.

1.2 Scope

The Virtual Whiteboard provides an infinite, zoomable canvas with drawing tools, shapes, text, sticky notes and image support, together with live collaboration, user accounts, board sharing, and export. It runs in modern web browsers on desktop, tablet and mobile devices.

In scope:

- User registration, authentication and profile management
- Creating, saving, organizing and deleting whiteboards
- Drawing, shape, text, sticky-note and image tools
- Real-time multi-user collaboration with presence and cursors
- Sharing with role-based permissions; commenting; version history
- Export to PNG, PDF and SVG

Out of scope (v1.0): built-in audio/video conferencing, native desktop apps, handwriting-to-text conversion, and AI-assisted diagramming.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification
Board	A single whiteboard document containing canvas objects
Canvas	The drawable, pannable and zoomable workspace of a board
Object	Any element on a canvas (stroke, shape, text, sticky note, image)
Owner / Editor / Viewer	Roles with decreasing permissions on a board
CRDT	Conflict-free Replicated Data Type used for merge-safe concurrent editing
WebSocket	Full-duplex protocol used for real-time updates
JWT	JSON Web Token used for session authentication

1.4 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications
- ISO/IEC/IEEE 29148:2018, Requirements Engineering
- WCAG 2.1, Web Content Accessibility Guidelines
- OWASP Application Security Verification Standard (ASVS)

1.5 Document Overview

Section 2 describes the product context. Section 3 lists functional requirements by feature. Sections 4 and 5 cover interfaces and quality attributes. Section 6 gives system models, and Sections 7–9 cover remaining requirements, traceability and appendices.

2. Overall Description

2.1 Product Perspective

The Virtual Whiteboard is a standalone, cloud-hosted web application built on a client–server architecture. The browser client renders the canvas and sends operations over WebSockets to a collaboration server, which broadcasts changes to other participants and persists board state in a database.

2.2 Product Functions (Summary)

- Account management and secure login
- Board creation, naming, templates and folders
- Freehand drawing, shapes, connectors, text, sticky notes, image upload
- Select, move, resize, rotate, group, layer and delete objects
- Undo/redo, infinite canvas, zoom and pan
- Real-time collaboration with live cursors and presence list
- Share via link or invitation with role-based access
- Comments and mentions; version history and restore
- Export and print

2.3 User Classes and Characteristics

User Class	Description	Technical Skill
Owner	Creates the board; manages sharing, roles and deletion	Low–Medium
Editor	Can add, modify and delete content and comment	Low
Viewer	Read-only access; may be allowed to comment	Low
Guest	Joins through a public link without an account (limited rights)	Low
Administrator	Manages users, abuse reports and system settings	High

2.4 Operating Environment

- Client: latest two versions of Chrome, Edge, Firefox and Safari; iOS 16+ and Android 12+ browsers
- Input: mouse, touch, stylus (pressure-sensitive where supported)
- Server: Linux-based cloud hosting, containerized deployment
- Database: relational database for metadata; object storage for images and exports

2.5 Design and Implementation Constraints

- Must use HTTPS/WSS for all communication
- Canvas rendering uses HTML5 Canvas or WebGL
- Real-time sync must use a conflict-resolution strategy (CRDT or Operational Transformation)
- Must comply with applicable privacy law (e.g., GDPR)

2.6 Assumptions and Dependencies

- Users have a stable internet connection (minimum 1 Mbps)
- Third-party services: email delivery (invitations, password reset), optional OAuth providers
- Browsers support WebSocket and the Canvas API

3. System Features and Functional Requirements

Priority: **H** = High (must have), **M** = Medium (should have), **L** = Low (nice to have).

3.1 User Account Management

ID	Requirement	Priority
FR-1.1	The system shall allow users to register using email and password.	H
FR-1.2	The system shall verify email addresses via a confirmation link.	H
FR-1.3	The system shall allow login with email/password and optionally Google or Microsoft OAuth.	M
FR-1.4	The system shall provide password reset via an expiring email link.	H
FR-1.5	The system shall allow users to edit their display name, avatar and cursor color.	L
FR-1.6	The system shall allow users to delete their account and associated personal data.	M

3.2 Board Management

ID	Requirement	Priority
FR-2.1	The system shall allow authenticated users to create a new blank board or one from a template.	H
FR-2.2	The system shall auto-save board changes without user action.	H
FR-2.3	The system shall list a user's boards with name, thumbnail, owner and last-modified date.	H
FR-2.4	The system shall allow renaming, duplicating, archiving, and deleting boards.	H
FR-2.5	The system shall support organizing boards into folders and marking favorites.	L
FR-2.6	The system shall provide search by board name.	M

3.3 Drawing and Canvas Tools

ID	Requirement	Priority
FR-3.1	The system shall provide a pen tool with selectable color, thickness and opacity.	H
FR-3.2	The system shall provide an eraser (stroke and object erase modes).	H
FR-3.3	The system shall provide shape tools: line, arrow, rectangle, ellipse, triangle and polygon.	H
FR-3.4	The system shall provide a text tool with font family, size, weight and alignment options.	H
FR-3.5	The system shall provide sticky notes with selectable colors and editable text.	M
FR-3.6	The system shall allow image upload (PNG, JPEG, GIF, SVG) by file picker, drag-and-drop or paste.	M
FR-3.7	The system shall provide connectors that remain attached to shapes when the shapes are moved.	M
FR-3.8	The system shall provide an infinite canvas with zoom (10%–400%) and pan.	H

ID	Requirement	Priority
FR-3.9	The system shall provide optional grid display and snap-to-grid.	L
FR-3.10	The system shall provide a laser pointer tool visible only temporarily to all participants.	L

3.4 Object Manipulation

ID	Requirement	Priority
FR-4.1	The system shall allow selecting single or multiple objects (click, shift-click, marquee).	H
FR-4.2	The system shall allow moving, resizing, rotating, copying, pasting and deleting objects.	H
FR-4.3	The system shall support grouping/ungrouping and bring-to-front/send-to-back layering.	M
FR-4.4	The system shall support lock/unlock of objects to prevent accidental edits.	L
FR-4.5	The system shall support undo and redo of at least the last 100 actions per user.	H

3.5 Real-Time Collaboration

ID	Requirement	Priority
FR-5.1	The system shall propagate any edit to all participants of the same board within 300 ms under normal conditions.	H
FR-5.2	The system shall display each participant's live cursor with name and color.	H
FR-5.3	The system shall display a list of current participants (presence).	M
FR-5.4	The system shall resolve concurrent edits without data loss using a CRDT or OT algorithm.	H
FR-5.5	The system shall allow offline edits to be queued and synchronized upon reconnection.	M
FR-5.6	The system shall support a 'follow user' mode that mirrors another participant's viewport.	L

3.6 Sharing and Permissions

ID	Requirement	Priority
FR-6.1	The system shall allow the owner to invite users by email and assign Editor or Viewer roles.	H
FR-6.2	The system shall allow generation of a shareable link with configurable access (none / view / edit).	H
FR-6.3	The system shall allow the owner to revoke access or change roles at any time, effective immediately.	H
FR-6.4	The system shall optionally protect shared links with a password and expiry date.	L
FR-6.5	The system shall enforce permissions on the server for every operation.	H

3.7 Comments and Communication

ID	Requirement	Priority
FR-7.1	The system shall allow users to attach comment threads to a location or object on the canvas.	M
FR-7.2	The system shall support @mentions that notify the mentioned user.	L
FR-7.3	The system shall allow comment threads to be resolved and re-opened.	L
FR-7.4	The system shall provide a lightweight in-board chat panel.	L

3.8 Version History

ID	Requirement	Priority
FR-8.1	The system shall automatically save board snapshots at regular intervals and on major changes.	M
FR-8.2	The system shall allow Owners and Editors to view and restore a previous version.	M
FR-8.3	The system shall display who made changes and when.	L

3.9 Export, Import and Print

ID	Requirement	Priority
FR-9.1	The system shall export the entire board or a selected area as PNG, PDF and SVG.	H
FR-9.2	The system shall allow export with a transparent or solid background.	L
FR-9.3	The system shall provide a print-friendly layout.	L

3.10 Administration

ID	Requirement	Priority
FR-10.1	The system shall provide an admin dashboard to view, suspend and delete user accounts.	M
FR-10.2	The system shall provide a mechanism to report abusive content and review reports.	M
FR-10.3	The system shall record audit logs of security-relevant events.	M

4. External Interface Requirements

4.1 User Interfaces

- **Main canvas view:** central canvas, left or bottom toolbar, top bar (board name, share, export, participants), right-side properties panel and comments panel.
- **Dashboard:** grid/list of boards, folders, search and 'New Board' action.
- **Responsive layout:** toolbar and panels collapse on screens narrower than 768 px.
- **Keyboard shortcuts:** V (select), P (pen), E (eraser), T (text), R (rectangle), Ctrl/Cmd+Z (undo), Ctrl/Cmd+Shift+Z (redo), Ctrl/Cmd+C/V (copy/paste), Space+drag (pan).
- **Accessibility:** all non-canvas controls reachable by keyboard with visible focus.

4.2 Hardware Interfaces

The application shall accept input from mouse, touchscreen, trackpad and stylus, including multi-touch gestures (pinch-to-zoom, two-finger pan) and pen pressure where the device exposes it.

4.3 Software Interfaces

Interface	Purpose
REST API (JSON over HTTPS)	Accounts, boards, sharing, comments, exports
WebSocket (WSS)	Real-time operations, presence and cursor updates
Relational database	Users, board metadata, permissions, comments
Object storage	Uploaded images, snapshots, exported files
Email service (SMTP/API)	Verification, invitations, password reset, notifications
OAuth 2.0 providers	Optional third-party sign-in

4.4 Communication Interfaces

All client–server traffic shall use TLS 1.2 or higher. WebSocket connections shall authenticate with a short-lived token and automatically reconnect with exponential back-off.

5. Non-Functional Requirements

5.1 Performance

ID	Requirement
NFR-P1	Initial board load shall complete within 3 seconds for boards of up to 5,000 objects on a broadband connection.
NFR-P2	Local drawing latency (input to render) shall not exceed 50 ms; canvas shall sustain at least 30 FPS during pan and zoom.
NFR-P3	Remote updates shall reach other participants within 300 ms at the 95th percentile.
NFR-P4	The system shall support at least 50 simultaneous editors per board and 10,000 concurrent users overall.

5.2 Security

ID	Requirement
NFR-S1	Passwords shall be stored using a salted, adaptive hash (bcrypt, scrypt or Argon2).
NFR-S2	Sessions shall use short-lived JWT access tokens with refresh-token rotation.
NFR-S3	All inputs shall be validated and sanitized to prevent XSS, CSRF, injection and SSRF attacks.
NFR-S4	Access control shall be enforced server-side on every REST and WebSocket operation.
NFR-S5	Data at rest shall be encrypted; uploaded files shall be scanned and restricted by type and size (max 10 MB).
NFR-S6	Login endpoints shall be rate-limited, and accounts temporarily locked after repeated failed attempts.

5.3 Reliability and Availability

ID	Requirement
NFR-R1	The service shall achieve at least 99.9% monthly uptime, excluding scheduled maintenance.
NFR-R2	No more than 1 second of work shall be lost in the event of a client crash or network drop.
NFR-R3	Data shall be backed up daily with a recovery point objective (RPO) of 1 hour and recovery time objective (RTO) of 4 hours.

5.4 Usability and Accessibility

ID	Requirement
NFR-U1	A new user shall be able to create a board and draw within 2 minutes without training.
NFR-U2	Non-canvas UI shall conform to WCAG 2.1 Level AA (contrast, keyboard operability, screen-reader labels).
NFR-U3	The interface shall support localization; English shall be the default language, with a framework for adding others.

5.5 Scalability, Maintainability and Portability

ID	Requirement
NFR-M1	Collaboration servers shall scale horizontally behind a load balancer.
NFR-M2	Code shall be modular, version-controlled and covered by automated unit tests (target ≥ 80% coverage on core logic).
NFR-M3	The system shall be deployable via containers and configured via environment variables.
NFR-M4	The application shall work consistently across the supported browsers and devices in Section 2.4.

6. System Models

6.1 Use Cases

ID	Use Case	Primary Actor	Related FRs
UC-1	Register and log in	Guest / User	FR-1.1 – 1.4
UC-2	Create and manage a board	Owner	FR-2.1 – 2.6
UC-3	Draw and edit content	Owner / Editor	FR-3.x, FR-4.x
UC-4	Collaborate in real time	Owner / Editor / Viewer	FR-5.x
UC-5	Share a board and manage roles	Owner	FR-6.x
UC-6	Comment on a board	Editor / Viewer	FR-7.x
UC-7	Restore a previous version	Owner / Editor	FR-8.x
UC-8	Export a board	Any authorized user	FR-9.x
UC-9	Moderate users and content	Administrator	FR-10.x

6.2 Detailed Use Case: Collaborate in Real Time (UC-4)

Field	Description
Actor	Editor (primary), other participants (secondary)
Preconditions	User is logged in or holds a valid share link; the board exists and the user has at least Viewer access.
Main flow	1. User opens the board. 2. Client establishes a WebSocket connection and loads the latest state. 3. Server adds the user to the presence list and notifies others. 4. User draws or edits; client applies the change locally and sends the operation. 5. Server validates permissions, merges the operation, persists it, and broadcasts it. 6. Other clients apply the update and render it.
Alternate flow	Connection lost: client keeps working offline, queues operations, and resyncs on reconnection.
Exception flow	Viewer attempts an edit: server rejects the operation and the client shows a read-only notice.
Postconditions	All participants converge on the same board state; changes are saved.

6.3 Logical Architecture (Overview)

Layer	Components
Client	Web UI, canvas renderer, tool manager, local state store, sync client
Application	REST API, authentication service, board service, sharing service, export service
Real-time	WebSocket gateway, collaboration engine (CRDT/OT), presence service
Data	Relational database, object storage, cache (for presence and sessions)
Infrastructure	Load balancer, container orchestration, monitoring and logging

6.4 Principal Data Entities

Entity	Key Attributes
User	userId, email, passwordHash, displayName, avatar, role, createdAt
Board	boardId, ownerId, title, thumbnail, createdAt, updatedAt, isArchived
BoardMember	boardId, userId, role (Owner/Editor/Viewer), invitedAt
ShareLink	linkId, boardId, accessLevel, passwordHash, expiresAt
CanvasObject	objectId, boardId, type, properties (JSON), zIndex, createdBy, updatedAt
Comment	commentId, boardId, objectId/position, authorId, text, parentId, resolved
Snapshot	snapshotId, boardId, dataRef, createdAt, createdBy

7. Other Requirements

7.1 Legal and Regulatory

- Provide a privacy policy and terms of service accepted at registration.
- Support data export and deletion on request (GDPR rights).
- Obtain consent for non-essential cookies and analytics.

7.2 Data Retention

- Deleted boards shall be recoverable for 30 days, then permanently purged.
- Audit logs shall be retained for at least 12 months.

7.3 Logging and Monitoring

The system shall log errors and key events, expose health-check endpoints, and alert operators when error rates or latency exceed defined thresholds.

7.4 Documentation

A user guide, API documentation and an administrator guide shall be delivered with the release.

8. Requirements Traceability

Business Goal	Supporting Requirements
Enable remote teams to brainstorm together	FR-5.x, FR-6.x, NFR-P3, NFR-P4
Offer an intuitive, fast drawing experience	FR-3.x, FR-4.x, NFR-P2, NFR-U1
Protect user data and content	FR-1.x, FR-6.5, NFR-S1 – S6, Section 7.1
Preserve and reuse work	FR-2.2, FR-8.x, FR-9.x, NFR-R2, NFR-R3
Be accessible to all users	NFR-U2, NFR-U3, Section 4.1

9. Appendices

Appendix A – Open Issues and Assumptions

- Final choice between CRDT and Operational Transformation to be confirmed during technical design.
- Maximum board size and per-user storage quota to be defined with stakeholders.
- Licensing/pricing tiers (free vs. paid) are not covered in v1.0.
- Need for audio/video integration to be reviewed for a future release.

Appendix B – Future Enhancements

- Native mobile and desktop apps
- Handwriting recognition and shape auto-correction
- AI-assisted summarization and diagram generation

- Integrations with Slack, Teams and Google Drive
- Presentation mode and templates marketplace

Appendix C – Revision History

Version	Date	Author	Description
0.1	2026-10-01	Project Team	Initial outline
1.0	2026-10-09	Project Team	First complete draft